

Envelope Equations

USPAS - Beam Physics with Intense Space Charge

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Paraxial ray equation

Envelope equations for axially symmetric beams

Envelope equations for elliptically symmetric beams

Envelope equations: reduced-order characterization of beam dynamics in terms of second-order moments

[Image]

Road map

- Start from single particle equation of motion: $\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$.
- Make use of:
 - Paraxial (near-axis) approximation
 - Conservation of angular momentum
 - Axisymmetry of external forces
- Arrive at single-particle paraxial ray equation.
- Take statistical averages to obtain moment/envelope equations.

Paraxial ray equation

Equation of motion in cylindrical coordinates

Equation of motion for particle of mass m and charge q in electric field $\mathbf{E}$ and magnetic field $\mathbf{B}$:

$$\frac{d\mathbf{p}}{dt} = \frac{d(\gamma m \dot{\mathbf{x}})}{dt} = q(\mathbf{E} + \dot{\mathbf{x}} \times \mathbf{B}), \quad (1)$$

Switch to cylindrical coordinates (r, θ, z) :

$$\begin{aligned} \mathbf{p} &= p_x \hat{x} + p_y \hat{y} + p_z \hat{z} \\ &= p_r \hat{r} + p_\theta^* \hat{\theta} + p_z \hat{z}. \end{aligned} \quad (2)$$

The components p_r and p_θ^* are given by

$$\begin{aligned} p_r &= \gamma m \dot{r}, \\ p_\theta^* &= \gamma m r \dot{\theta}. \end{aligned} \quad (3)$$

The symbol p_θ^* is used to reserve p_θ for the *canonical* angular momentum, defined later.

In cylindrical coordinates, the unit vectors depend on the particle position and thus change with time:

$$\begin{aligned}\dot{\hat{r}} &\equiv d\hat{r}/dt = \dot{\theta}\hat{\theta}, \\ \dot{\hat{\theta}} &\equiv d\hat{\theta}/dt = -\dot{\theta}\hat{r}.\end{aligned}\tag{4}$$

Thus, the time derivative of the momentum vector in Eq. (2) is:

$$\frac{d\mathbf{p}}{dt} = \frac{d}{dt} (p_r\hat{r} + p_\theta^*\hat{\theta} + p_z\hat{z})\tag{5}$$

$$= \dot{p}_r\hat{r} + p_r\dot{\hat{r}} + \dot{p}_\theta\hat{\theta} + p_\theta^*\dot{\hat{\theta}} + \dot{p}_z\hat{z}\tag{6}$$

$$= (\dot{p}_r - p_\theta^*\dot{\theta})\hat{r} + (\dot{p}_\theta + p_r\dot{\theta})\hat{\theta} + \dot{p}_z\hat{z}\tag{7}$$

$$= \left[\frac{d(\gamma m \dot{r})}{dt} - \gamma m r \dot{\theta}^2 \right] \hat{r} + \left[\frac{d(\gamma m r \dot{\theta})}{dt} + \gamma m \dot{r} \dot{\theta} \right] \hat{\theta} + \left[\frac{d(\gamma m \dot{z})}{dt} \right] \hat{z}.\tag{8}$$

$$= \left[\frac{d(\gamma m \dot{r})}{dt} - \gamma m r \dot{\theta}^2 \right] \hat{r} + \left[\frac{1}{r} \frac{d}{dt} (\gamma m r^2 \dot{\theta}) \right] \hat{\theta} + \left[\frac{d(\gamma m \dot{z})}{dt} \right] \hat{z}.\tag{9}$$

This takes care of the left side of the Lorentz force equation (Eq. (1)).

The right side of Eq. (1) requires the cylindrical field components:

$$\begin{aligned}\mathbf{E} &= E_r \hat{r} + E_\theta \hat{\theta} + E_z \hat{z}, \\ \mathbf{B} &= B_r \hat{r} + B_\theta \hat{\theta} + B_z \hat{z}.\end{aligned}\tag{10}$$

Evaluating the cross product $\dot{\mathbf{x}} \times \mathbf{B}$,

$$\dot{\mathbf{x}} \times \mathbf{B} = \begin{vmatrix} \hat{r} & \hat{\theta} & \hat{z} \\ \dot{r} & r\dot{\theta} & \dot{z} \\ B_r & B_\theta & B_z \end{vmatrix},\tag{11}$$

leads to

$$\frac{d(\gamma m \dot{r})}{dt} - \gamma m r \dot{\theta}^2 = q (E_r + r \dot{\theta} B_z - \dot{z} B_\theta),\tag{12}$$

$$\frac{1}{r} \frac{d(\gamma m r^2 \dot{\theta})}{dt} = q (E_\theta + \dot{z} B_r - \dot{r} B_z),\tag{13}$$

$$\frac{d(\gamma m \dot{z})}{dt} = q (E_z + \dot{r} B_\theta - r \dot{\theta} B_r)\tag{14}$$

From fields to potentials

Write the fields in terms of the scalar potential $\phi(\mathbf{x})$ and vector potential $\mathbf{A}(\mathbf{x})$:

$$\begin{aligned}\mathbf{E} &= -\nabla\phi - \frac{\partial\mathbf{A}}{\partial t}, \\ \mathbf{B} &= \nabla \times \mathbf{A}.\end{aligned}\tag{15}$$

The gradient $\nabla\phi$ in cylindrical coordinates is:

$$\nabla\phi = \left[\frac{\partial\phi}{\partial r}\right]\hat{r} + \left[\frac{1}{r}\frac{\partial\phi}{\partial\theta}\right]\hat{\theta} + \left[\frac{\partial\phi}{\partial z}\right]\hat{z}.\tag{16}$$

And the curl $\nabla \times \mathbf{A}$ is:

$$\nabla \times \mathbf{A} = \left[\frac{1}{r}\frac{\partial A_z}{\partial\theta} - \frac{\partial A_\theta}{\partial z}\right]\hat{r} + \left[\frac{\partial A_r}{\partial z} - \frac{\partial A_z}{\partial r}\right]\hat{\theta} + \frac{1}{r}\left[\frac{\partial(rA_\theta)}{\partial r} - \frac{\partial A_r}{\partial\theta}\right]\hat{z}.\tag{17}$$

Axial symmetry means all partial derivatives with respect to θ are zero in Eq. (16) and Eq. (17). This gives the following expressions for the fields in terms of the potentials in cylindrical coordinates:

$$\begin{aligned}\mathbf{E} &= \left[-\frac{\partial\phi}{\partial r} - \frac{\partial A_r}{\partial t} \right] \hat{r} + \left[-\frac{\partial A_\theta}{\partial t} \right] \hat{\theta} + \left[-\frac{\partial\phi}{\partial z} - \frac{\partial A_z}{\partial t} \right] \hat{z}, \\ \mathbf{B} &= \left[-\frac{\partial A_\theta}{\partial r} \right] \hat{r} + \left[\frac{\partial A_r}{\partial z} - \frac{\partial A_z}{\partial r} \right] \hat{\theta} + \left[\frac{1}{r} \frac{\partial(rA_\theta)}{\partial r} \right] \hat{z}.\end{aligned}\tag{18}$$

Conservation of angular momentum

Use the potentials to rewrite the angular equation of motion (Eq. (13)):

$$\begin{aligned}\frac{d}{dt}(\gamma m r^2 \dot{\theta}) &= q r (E_{\theta} + \dot{z} B_r - \dot{r} B_z) \\&= q r \left(\left[-\frac{\partial A_{\theta}}{\partial t} \right] + \dot{z} \left[-\frac{\partial A_{\theta}}{\partial z} \right] - \dot{r} \left[\frac{1}{r} \frac{\partial(r A_{\theta})}{\partial r} \right] \right) \\&= -q \left(\frac{\partial(r A_{\theta})}{\partial t} + \dot{z} \frac{\partial(r A_{\theta})}{\partial z} + \dot{r} \frac{\partial(r A_{\theta})}{\partial r} \right) \\&= -q \left[\frac{\partial}{\partial t} + \dot{z} \frac{\partial}{\partial z} + \dot{r} \frac{\partial}{\partial r} \right] (r A_{\theta}) \\&= -q \left[\frac{\partial}{\partial t} + \dot{\mathbf{x}} \cdot \nabla \right] (r A_{\theta}) \\&= -\frac{d(q r A_{\theta})}{dt}.\end{aligned}\tag{19}$$

Eq. (19) can then be written as:

$$\frac{d}{dt} (\gamma m r^2 \dot{\theta} + q r A_{\theta}) = 0. \quad (20)$$

This motivates the definition of the *canonical angular momentum*:

$$p_{\theta} \equiv \gamma m r^2 \dot{\theta} + q r A_{\theta}, \quad (21)$$

such that $dp_{\theta}/dt = 0$. We may also write p_{θ} in terms of the magnetic flux $\psi(r)$ though a circle of radius r [Figure]:

$$\psi(r) = \int \mathbf{B} \cdot d\Omega = \int \nabla \times \mathbf{A} \cdot d\Omega = \oint \mathbf{A} \cdot d\mathbf{l} = 2\pi r A_{\theta}. \quad (22)$$

Therefore, Eq. (21) can be written

$$p_{\theta} \equiv \gamma m r^2 \dot{\theta} + \frac{q\psi(r)}{2\pi}. \quad (23)$$

Series expansion of axially symmetric fields

Consider external fields $\mathbf{E}$ and $\mathbf{B}$ with azimuthal symmetry ($\partial/\partial\theta = 0$). In the absence of charge and current densities,

$$\begin{aligned}\nabla \cdot \mathbf{E} &= 0, \\ \nabla \cdot \mathbf{B} &= 0, \\ \nabla \times \mathbf{E} &= 0, \\ \nabla \times \mathbf{B} &= 0.\end{aligned}\tag{24}$$

The vanishing curl means we can express the fields as

$$\begin{aligned}\mathbf{E} &= -\nabla\phi, \\ \mathbf{B} &= -\nabla\phi_m,\end{aligned}\tag{25}$$

for scalar potentials ϕ and ϕ_m . Additionally, the vanishing divergence leads to the Laplace equation for the scalar potential:

$$\nabla^2\phi = 0.\tag{26}$$

In cylindrical coordinates, the Laplace equation for the potential is:

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial \phi}{\partial r} \right) + \frac{\partial^2 \phi}{\partial z^2} = 0. \quad (27)$$

The azimuthal symmetry will lead to a simplified form of the scalar potential in terms of its on-axis derivatives along the z axis. First, expand the potential $\phi(r, z)$ as a power series,

$$\phi(r, z) = \sum_{n=0}^{\infty} h_{2n}(z) r^{2n}, \quad (28)$$

where the $h_{2n}(z)$ are functions of z . The on-axis potential is given by $h_0(z) = \phi(0, z)$.

The rotational symmetry eliminates all odd powers of r in the series.

At the origin ($r = 0$), we must have $E_r = B_r = 0$. If we allowed the $n = 1$ term $h_1(z)r$, we would have $E_r(0, z) = -\partial\phi(r, z)/\partial r|_{r=0} = h_1(z) \neq 0$.

The derivatives in the Laplace equation are:

$$\begin{aligned}\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial \phi}{\partial r} \right) &= \sum_{n=1}^{\infty} (2n)^2 h_{2n}(z) r^{2n-2}, \\ \frac{\partial^2 \phi}{\partial z^2} &= \sum_{n=0}^{\infty} h''_{2n}(z) r^{2n},\end{aligned}\tag{29}$$

where $h'(z) = \partial h(z) / \partial z$.

Thus, $\nabla^2\phi = 0$ becomes:

$$\begin{aligned}
\sum_{n=1}^{\infty} (2n)^2 h_{2n}(z) r^{2n-2} + \sum_{n=0}^{\infty} h_{2n}''(z) r^{2n} &= 0 \\
\sum_{n=0}^{\infty} (2n+2)^2 h_{2n+2}(z) r^{2n} + \sum_{n=0}^{\infty} h_{2n}''(z) r^{2n} &= 0 \\
\sum_{n=0}^{\infty} [(2n+2)^2 h_{2n+2}(z) + h_{2n}''(z)] &= 0
\end{aligned} \tag{30}$$

The result is a recurrence relation,

$$h_{2n+2}(z) = -\frac{h_{2n}''(z)}{(2n+2)^2}, \tag{31}$$

with which we can write all terms as derivatives of $h_0(z)$ with respect to z :

$$\begin{aligned}
h_2(z) &= -\frac{1}{4} \frac{\partial^2 h_0(z)}{\partial z^2}, \\
h_4(z) &= +\frac{1}{64} \frac{\partial^4 h_0(z)}{\partial z^4}, \\
&\vdots
\end{aligned} \tag{32}$$

We may thus write the potential at all points in space using only the z -derivatives of the on-axis potential $\phi(0, z)$:

$$\phi(r, z) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(n!)^2} \frac{\partial^{2n} \phi(0, z)}{\partial z^{2n}} \left(\frac{r}{2}\right)^{2n}. \quad (33)$$

We now express the fields in terms of the expanded scalar potentials. Write the magnetic field as $\mathbf{B} = B_r \hat{r} + B_z \hat{z}$. Using the series expansion of $\phi_m(r, z)$, we find for the z component of the field:

$$\begin{aligned}
B_z(r, z) &= -\frac{\partial \phi_m}{\partial z} \\
&= -\frac{\partial}{\partial z} \left[\sum_{n=0}^{\infty} \frac{(-1)^{2n}}{(n!)^2} \frac{\partial^{2n} \phi_m(0, z)}{\partial z^{2n}} \left(\frac{r}{2}\right)^2 \right] \\
&= -\sum_{n=0}^{\infty} \frac{(-1)^{2n}}{(n!)^2} \frac{\partial^{2n+1} \phi_m(0, z)}{\partial z^{2n+1}} \left(\frac{r}{2}\right)^2.
\end{aligned} \tag{34}$$

Define $B(z)$ as the on-axis value of $B_z(z)$:

$$B(z) \equiv B_z(0, z) = -\frac{\partial \phi_m(0, z)}{\partial z}. \tag{35}$$

Inserting Eq. (35) into Eq. (34) gives:

$$\begin{aligned}
B_z(r, z) &= \sum_{n=0}^{\infty} \frac{(-1)^{2n}}{(n!)^2} \frac{\partial^{2n}}{\partial z^{2n}} \left[-\frac{\partial \phi_m(0, z)}{\partial z} \right] \left(\frac{r}{2} \right)^2 \\
B_z(r, z) &= \sum_{n=0}^{\infty} \frac{(-1)^{2n}}{(n!)^2} \frac{\partial^{2n} B(z)}{\partial z^{2n}} \left(\frac{r}{2} \right)^2.
\end{aligned} \tag{36}$$

For the radial component, we find:

$$B_r(r, z) = \sum_{n=1}^{\infty} \frac{(-1)^n}{n!(n-1)!} \frac{\partial^{2n-1} B(z)}{\partial z^{2n-1}} \left(\frac{r}{2} \right)^{2n-1}. \tag{37}$$

A similar procedure gives the radial and longitudinal components of the electric field. Define $V(z)$ as the on-axis value of the electric potential ϕ :

$$V(z) \equiv \phi(0, z). \quad (38)$$

This gives:

$$\begin{aligned} E_z(r, z) &= -\frac{\partial \phi}{\partial z} = -\sum_{n=0}^{\infty} \frac{(-1)^n}{(n!)^2} \frac{\partial^{2n+1} V(z)}{\partial z^{2n+1}} \left(\frac{r}{2}\right)^{2n}. \\ E_r(r, z) &= -\frac{\partial \phi}{\partial r} = -\sum_{n=1}^{\infty} \frac{(-1)^n}{n!(n-1)!} \frac{\partial^{2n} V(z)}{\partial z^{2n}} \left(\frac{r}{2}\right)^{2n-1}. \end{aligned} \quad (39)$$

Derivation of paraxial ray equation

We will now take the *paraxial* limit, assuming small transverse position and velocity: $\dot{r} \ll \dot{z}$, $r\dot{\theta} \ll \dot{z}$. If $v = (\dot{r}^2 + r^2\dot{\theta}^2 + \dot{z}^2)^{1/2}$ is the total velocity, the paraxial limit gives $\dot{z} = (v^2 - \dot{r}^2 - r^2\dot{\theta}^2)^{1/2} \approx v$.

To first order in r and r' , the external fields are:

$$\begin{aligned} E_r^{(ext)} &\approx \frac{1}{2} \frac{\partial^2 V(z)}{\partial z^2} r, \\ B_z^{(ext)} &\approx B(z), \\ B_\theta^{(ext)} &= 0 (\partial\phi_m/\partial\theta = 0). \end{aligned} \tag{40}$$

Plug these truncated fields into the radial equation of motion (Eq. (12)):

$$\begin{aligned} \frac{d(\gamma m \dot{r})}{dt} - \gamma m r \dot{\theta}^2 &= q (E_r + r \dot{\theta} B_z - \dot{z} B_\theta), \\ &\approx q \left(\frac{1}{2} \frac{\partial^2 V(z)}{\partial z^2} r + r \dot{\theta} B(z) \right) \end{aligned} \tag{41}$$

Two more steps. First, use z as independent variable: $dz = \beta c dt$. Using $' \equiv d/dz$ and dropping the $\approx$ symbol gives:

$$\beta c \frac{d(\gamma m \beta c r')}{dz} - \gamma m r (\beta c \theta')^2 = q \left(\frac{1}{2} V''(z) r + r \beta c \theta' B(z) \right). \quad (42)$$

Expand the derivative in the first term and divide by $\gamma \beta^2 m c^2$:

$$r'' + \frac{(\gamma \beta)'}{(\gamma \beta)} r' - r \theta'^2 = \frac{q}{\gamma \beta^2 m c^2} \left(\frac{1}{2} V''(z) + \beta c \theta' B(z) \right) r. \quad (43)$$

Now use the definition of the canonical angular momentum p_θ (Eq. (21)) to eliminate θ' , noting that the flux $\psi(r) = \pi r^2 B$:

$$\begin{aligned}
 \theta' &= \frac{\dot{\theta}}{\beta c} \\
 &= \frac{1}{\beta c} \frac{1}{\gamma m r^2} \left(p_\theta - \frac{q\psi(r)}{2\pi} \right), \\
 &= \frac{p_\theta}{\gamma \beta m c r^2} - \frac{\omega_c}{2\gamma \beta c},
 \end{aligned} \tag{44}$$

where we have defined the *cyclotron frequency*:

$$\omega_c \equiv \frac{qB}{m}. \tag{45}$$

Substitution of Eq. (44) into Eq. (43) gives:

$$r'' + \frac{(\gamma\beta)'}{\gamma\beta} r + \left(\frac{\omega_c}{2\gamma\beta c} \right)^2 r - \left(\frac{p_\theta}{\gamma\beta m c} \right)^2 \frac{1}{r^3} = \frac{q}{\gamma m \beta^2 c^2} \frac{1}{2} V''(z) r \tag{46}$$

The rate of change in the particle energy is given by

$$\frac{d(\gamma mc^2)}{dt} = q\mathbf{E} \cdot \mathbf{v}. \quad (47)$$

Rearrange and switch to z as independent variable. In the paraxial limit:

$$\gamma' = \frac{q}{mc^2} \frac{\mathbf{E} \cdot \mathbf{v}}{v_z} \quad (48)$$

$$= \frac{q}{mc^2} \frac{E_x v_x + E_y v_y + E_z v_z}{v_z} \quad (49)$$

$$\approx \frac{q}{mc^2} E_z. \quad (50)$$

Giving the approximate relationship:

$$\gamma'' \approx \frac{q}{mc^2} E'_z = -\frac{q}{mc^2} V''. \quad (51)$$

This approximation of λ'' leads to the *paraxial ray equation*:

$$\boxed{r'' + \frac{(\gamma\beta)'}{(\gamma\beta)}r' + \left(\frac{\gamma''}{2\gamma\beta^2}\right)r + \left(\frac{\omega_c}{2\gamma\beta c}\right)^2 r - \left(\frac{p_\theta}{\gamma\beta mc}\right)^2 \frac{1}{r^3} = 0.} \quad (52)$$

- (i) inertial term; (ii) accelerative damping; E_r from converging field lines;
- (iii) solenoidal focusing ($v_\theta B_z$, part of centrifugal term); (iv) centrifugal defocusing.

We can also account for linearized self-fields. Assuming cylindrical symmetry of the beam, the Poisson equation for the electric field is:

$$\nabla^2 \phi^{(self)} = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial \phi^{(self)}}{\partial r} \right) + \frac{\partial^2 \phi^{(self)}}{\partial s^2} = -\frac{\rho}{\epsilon_0}. \quad (53)$$

Solving for $\partial \phi^{(self)} / \partial r$:

$$E_r^{(self)} = -\frac{\partial \phi^{(self)}}{\partial r} = \frac{\lambda(r)}{2\pi\epsilon_0 r} + \frac{r}{2} \frac{\partial^2 \phi^{(self)}}{\partial s^2}, \quad (54)$$

where λ is the line-charge density:

$$\lambda(r) \equiv 2\pi \int_0^r \rho(\tilde{r}) \tilde{r} d\tilde{r}, \quad (55)$$

For the $B_{\theta}^{(self)}$ component of the self-generated magnetic field, we use Ampere's law:

$$2\pi r B_{\theta}^{(self)} = \mu_0 \int_0^r 2\pi \tilde{r} J_z(\tilde{r}) d\tilde{r} = \mu_0 \beta c \lambda(r) \quad (56)$$

Using the definition $\mu_0 \epsilon_0 = c^{-2}$,

$$B_{\theta}^{(self)} = \frac{\beta}{c} \frac{\lambda(r)}{2\pi \epsilon_0 r}. \quad (57)$$

Including the linear self fields in the paraxial limit adds a driving term to the right hand side:

$$\boxed{r'' + \frac{(\gamma\beta)'}{(\gamma\beta)} r' + \left(\frac{\gamma''}{2\gamma\beta^2} \right) r + \left(\frac{\omega_c}{2\gamma\beta c} \right)^2 r - \left(\frac{p_\theta}{\gamma\beta mc} \right)^2 \frac{1}{r^3} = \frac{q}{mc^2\beta^2\gamma^3} \frac{\lambda(r)}{2\pi\epsilon_0 r}}$$

(58)

Envelope equations for axially symmetric beams

The strategy in this section will be to compute the second moments of the transverse distribution function $f(x, x', y, y', s)$. For axially symmetric beams, the Vlasov equation will generate a set of differential equations for the evolution of the second moments such as $\langle rr \rangle$, $\langle rr' \rangle$, $\langle rr'' \rangle$, etc. Combining with the paraxial ray equation will generate a set of ordinary differential equations (ODEs) for the evolution of the beam *envelope* radius $r_b = \sqrt{2\langle rr \rangle}$.

Start with the Vlasov equation:

$$\frac{\partial f}{\partial s} + \mathbf{x}' \cdot \frac{\partial f}{\partial \mathbf{x}} + \mathbf{x}'' \cdot \frac{\partial f}{\partial \mathbf{x}'} = 0, \quad (59)$$

where $\mathbf{x} = (x, y)$.

Multiply the Vlasov equation by $g(\mathbf{x}, \mathbf{x}')$, an arbitrary function of the phase space coordinates, and integrate over all phase space space:

$$\iint \left[g \frac{\partial f}{\partial s} + g \mathbf{x}' \cdot \frac{\partial f}{\partial \mathbf{x}} + g \mathbf{x}'' \cdot \frac{\partial f}{\partial \mathbf{x}'} \right] d\mathbf{x} d\mathbf{x}' = 0. \quad (60)$$

Introduce bracket notation to indicate the expected value under the distribution; e.g.,

$$\langle g \rangle \equiv \iint g f(\mathbf{x}, \mathbf{x}') d\mathbf{x} d\mathbf{x}'. \quad (61)$$

The first term is equivalent to $d\langle g \rangle/ds$:

$$\iint g(\mathbf{x}, \mathbf{x}') \frac{\partial f(\mathbf{x}, \mathbf{x}', s)}{\partial s} d\mathbf{x} d\mathbf{x}' = \frac{d\langle g \rangle}{ds}. \quad (62)$$

[Shouldn't this be $\partial\langle g \rangle/\partial s$ instead of $d\langle g \rangle/ds$?]

Since the distribution function vanishes at the integration boundary, we can use integration by parts to flip the gradients from f to g in the next two terms.

$$\begin{aligned}\iint \left[g \mathbf{x}' \cdot \frac{\partial f}{\partial \mathbf{x}} \right] d\mathbf{x} d\mathbf{x}' &= - \iint \left[\mathbf{x}' \cdot \frac{\partial g}{\partial \mathbf{x}} f \right] d\mathbf{x} d\mathbf{x}' \\ &= -\langle \mathbf{x}' \cdot \partial g / \partial \mathbf{x} \rangle.\end{aligned}\tag{63}$$

$$\begin{aligned}\iint \left[g \mathbf{x}'' \cdot \frac{\partial f}{\partial \mathbf{x}'} \right] d\mathbf{x} d\mathbf{x}' &= - \iint \left[\mathbf{x}'' \cdot \frac{\partial g}{\partial \mathbf{x}'} f \right] d\mathbf{x} d\mathbf{x}' \\ &= -\langle \mathbf{x}'' \cdot \partial g / \partial \mathbf{x}' \rangle.\end{aligned}\tag{64}$$

Eq. (60) then becomes:

$$\frac{d\langle g \rangle}{ds} = \langle \mathbf{x}' \cdot \partial g / \partial \mathbf{x} \rangle + \langle \mathbf{x}'' \cdot \partial g / \partial \mathbf{x}' \rangle. \quad (65)$$

From the definition of the total derivative with respect to s :

$$\frac{dg}{ds} = \mathbf{x}' \cdot \partial g / \partial \mathbf{x} + \mathbf{x}'' \cdot \partial g / \partial \mathbf{x}'. \quad (66)$$

Therefore, the time-derivative of the average $\langle g \rangle$ is the same as the average of the time-derivative dg/dt :

$$d\langle g \rangle / ds = \langle g' \rangle. \quad (67)$$

[Is this not trivial?]

Define the root-mean-square (rms) beam radius r_b :

$$r_b \equiv \sqrt{2\langle rr \rangle}. \quad (68)$$

We will also call r_b the *envelope radius*. Eq. (68), the time-evolution of the rms beam radius is:

$$r_b'' = \frac{2\langle rr'' \rangle}{r_b} + \frac{4\langle rr \rangle \langle r' r' \rangle - 4\langle rr' \rangle^2}{r_b^3}. \quad (69)$$

Evaluate $\langle rr'' \rangle$: [...]

$$\begin{aligned} \langle rr'' \rangle = & \frac{\gamma'}{\beta^2 \gamma} \langle rr' \rangle + \frac{\gamma''}{2\gamma\beta^2} \langle rr \rangle - \frac{q}{mc^2 \beta^2 \gamma^3} \frac{\langle \lambda(r) \rangle}{2\pi\epsilon_0} \\ & + \left(\frac{\langle p_\theta \rangle}{\gamma\beta mc} \right)^2 \frac{1}{\langle r^2 \rangle} - \left(\frac{\omega_c}{2\gamma\beta c} \right)^2 \langle rr \rangle - \frac{\langle r^2 \theta' \rangle^2}{\langle rr \rangle} + \langle r^2 \theta'^2 \rangle. \quad (70) \end{aligned}$$

Inserting Eq. (??) Eq. (69) gives the following *rms envelope equation*: [...]

$$\boxed{r_b'' + \left(\frac{\gamma'}{\gamma\beta^2} \right) r_b' + \left(\frac{\gamma''}{2\gamma\beta^2} \right) r_b + \left(\frac{\omega_c}{2\gamma\beta c} \right) r_b - \left(\frac{2\langle p_\theta \rangle}{\gamma\beta mc} \right)^2 \frac{1}{r_b^3} - \frac{\varepsilon_r^2}{r_b^3} - \frac{Q}{r_b}}, \quad (71)$$

where Q is the *perveance*:

$$Q \equiv [...], \quad (72)$$

and ε_r is the *radial emittance*:

$$\begin{aligned} \varepsilon_r^2 &\equiv 4 \left(\langle r^2 \rangle \langle r'^2 \rangle - \langle rr' \rangle^2 + \langle r^2 \rangle \langle r^2 \theta'^2 \rangle - \langle r^2 \theta' \rangle^2 \right) \\ \varepsilon_r^2 &= \varepsilon_x^2 - 4 \langle r^2 \theta' \rangle^2. \end{aligned} \quad (73)$$

Envelope equations for elliptically symmetric beams
